Practice workbook · Relate sperm structures to their roles in fertilisation

Labelling sperm parts and explaining the roles of the tail, mitochondria-packed midpiece, acrosome and haploid nucleus.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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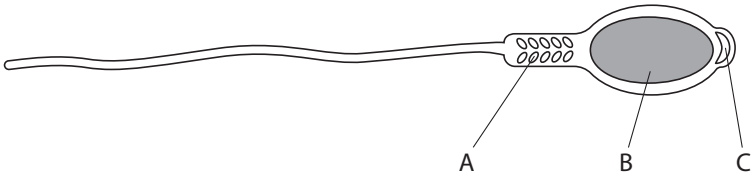
Type ID: HBI-RELATE-SPERM-STRUCTURE-FUNCTION

1. Question 6 · part a.ii · 3 marks

November 2023 Paper 2

Worked-example retry · Attempt part a.ii only.

6 The diagram shows a human sperm.



(a) (i) Name the parts A, B and C.

(3)

A .....

B .....

C .....

(ii) Describe the function of part C.

(3)

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(iii) The actual length of the sperm is 50.0 micrometres (μm).

Calculate the magnification of the diagram.

[1 mm = 1000 μm]

(4)

magnification = ...

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1. Question 6 · part a.ii · 3 marks · Working and answer

Working and answer

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2. Question 5 · part a.i · 2 marks

June 2021 Paper 1

Further practice · Attempt part a.i only.

5 The diagram shows a sperm.



(a) (i) Describe the function of a sperm.

(2)

(ii) State two other structures, apart from mitochondria, that can be seen in the sperm if it is viewed using an electron microscope.

(2)

1 .....

2 .....

(b) Explain why there are a large number of mitochondria found in a sperm.

(3)

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2. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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3. Question 5 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

5 The diagram shows a sperm.



(a) (i) Describe the function of a sperm.

(2)

(ii) State two other structures, apart from mitochondria, that can be seen in the sperm if it is viewed using an electron microscope.

(2)

1 .....

2 .....

(b) Explain why there are a large number of mitochondria found in a sperm.

(3)

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3. Question 5 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

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(c) The actual length of the sperm between X and Y on the diagram is 3 µm.

Calculate the magnification of the sperm shown in the diagram.

[1000 µm = 1 mm]

(4)

magnification = × .....

(Total for Question 5 = 11 marks)

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Working and answer

4. Question 2 · part a.i · 1 mark

January 2023 Paper 1

Further practice · Attempt part a.i only.

2 (a) A human sperm cell has a tail, a mid-piece containing organelles and a head with a nucleus.

(i) Give the function of the tail.

(1)

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(ii) Which of these, found in the mid-piece, release energy?

(1)

- ☒ **A** chromosomes
- ☒ **B** endoplasmic reticulum
- ☒ **C** mitochondria
- ☒ **D** ribosomes

(iii) The process of meiosis is involved in the production of sperm.

Explain the importance of meiosis in sperm production.

(3)

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4. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

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5. Question 2 · part b.ii · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.ii only.

2 (a) A human sperm cell has a tail, a mid-piece containing organelles and a head with a nucleus.

(i) Give the function of the tail.

(1)

(ii) Which of these, found in the mid-piece, release energy?

(1)

- ☒ **A** chromosomes
- ☒ **B** endoplasmic reticulum
- ☒ **C** mitochondria
- ☒ **D** ribosomes

(iii) The process of meiosis is involved in the production of sperm.

Explain the importance of meiosis in sperm production.

(3)

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5. Question 2 · part b.ii · 2 marks

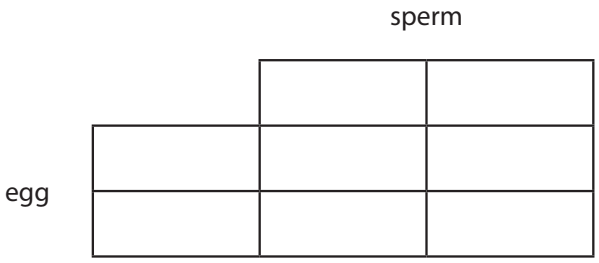
January 2023 Paper 1

Further practice · Attempt part b.ii only.

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(b) (i) Complete the genetic diagram to show how the sex of a person is determined by the chromosomes in the nucleus of the egg and the nucleus of the sperm.

(3)



(ii) The sperm head has to enter the egg cell so the nuclei can fuse.

Describe how the sperm head enters the egg cell.

(2)

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(Total for Question 2 = 10 marks)

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Working and answer

6. Question 3 · part b.iii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.iii only.

3 (a) Describe what is meant by a specialised cell.

(2)

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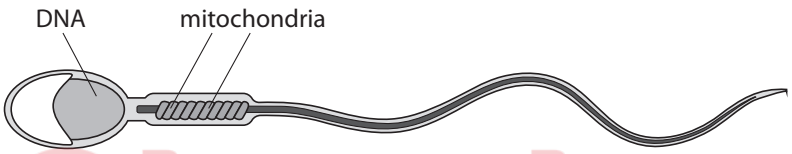
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(b) The sperm cell is an example of a specialised cell.

The diagram shows a sperm cell.

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(i) The sperm cell in the diagram has an actual length of 0.0060 mm.

The length of the image of the sperm cell is 84 mm.

Calculate the magnification of the sperm cell.

(2)

magnification = × .....

(ii) The head of the sperm contains DNA.

Give the difference in the amount of DNA found in the head of the sperm and in a nucleus of a body cell.

(1)

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6. Question 3 · part b.iii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.iii only.

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(iii) Explain why the sperm cell has many mitochondria.

(2)

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(iv) A sperm cell travels at a speed of 0.070 mm/s.

It travels a distance of 180 mm to fertilise an egg.

Use this equation to calculate the time it takes for the sperm to reach the egg.

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Give your answer in minutes.

(3)

time taken = ... minutes

(c) An electron microscope can be used to view structures in specialised cells.

Explain why an electron microscope is more useful than a light microscope to view cell structure.

(2)

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(Total for Question 3 = 12 marks)

6. Question 3 · part b.iii · 2 marks · Working and answer

Working and answer

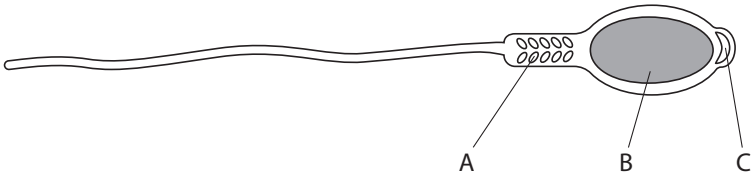
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7. Question 6 · part a.i · 3 marks

November 2023 Paper 2

Further practice · Attempt part a.i only.

6 The diagram shows a human sperm.



(a) (i) Name the parts A, B and C.

(3)

A .....

B .....

C .....

(ii) Describe the function of part C.

(3)

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(iii) The actual length of the sperm is 50.0 micrometres (μm).

Calculate the magnification of the diagram.

[1 mm = 1000 μm]

(4)

magnification = ...

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P 7 4 4 9 4 A 0 1 4 2 0

7. Question 6 · part a.i · 3 marks · Working and answer

Working and answer

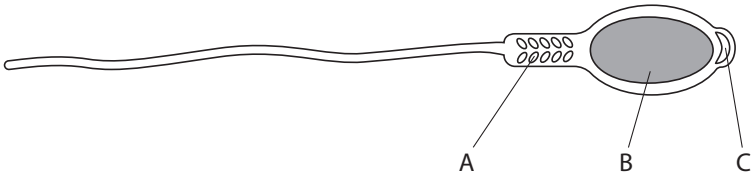
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8. Question 6 · part b · 3 marks

November 2023 Paper 2

Further practice · Attempt part b only.

6 The diagram shows a human sperm.



(a) (i) Name the parts A, B and C.

(3)

A .....

B .....

C .....

(ii) Describe the function of part C.

(3)

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(iii) The actual length of the sperm is 50.0 micrometres (μm).

Calculate the magnification of the diagram.

[1 mm = 1000 μm]

(4)

magnification = ...

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P 7 4 4 9 4 A 0 1 4 2 0

8. Question 6 · part b · 3 marks

November 2023 Paper 2

Further practice · Attempt part b only.

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(b) Describe the function of the structure labelled A in the process of fertilisation.

(3)

(Total for Question 6 = 13 marks)

Working and answer

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